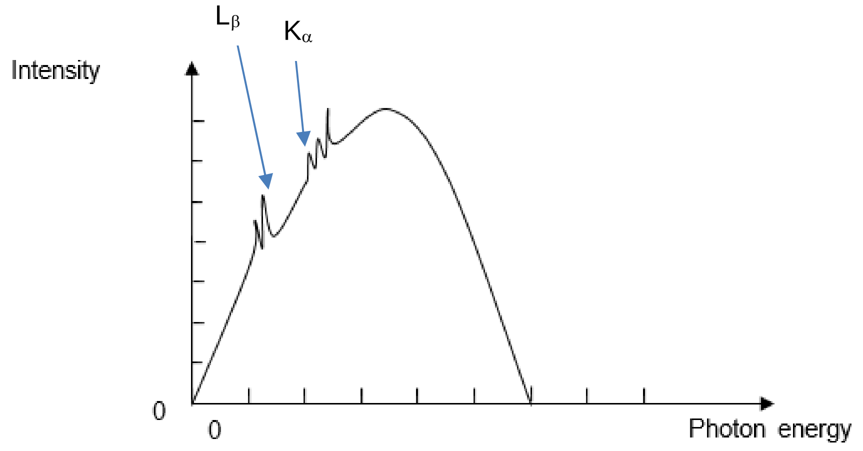
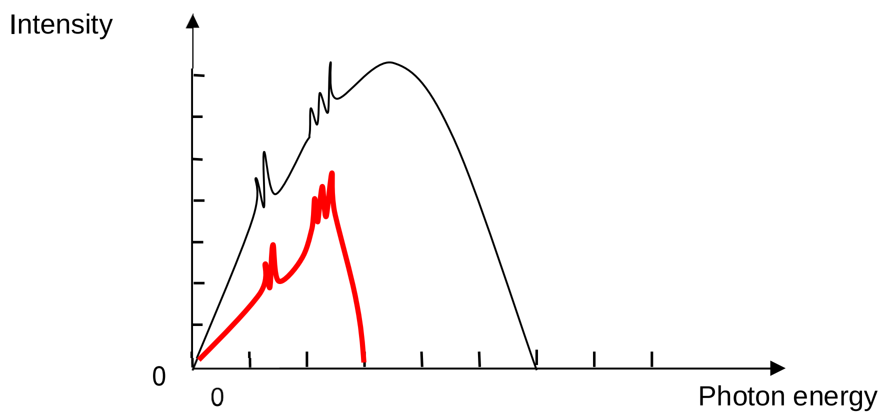
5	**(a)**	**(i)**	Work function is defined as the minimum energy required to eject an electron from the surface of a metal.	
		(ii)	Using loss in EPE = gain in E_k, $0.5mv^2 = eV_s$ $0.5(9.11 \times 10^{-31})v^2 = (1.6 \times 10^{-19})(1)$ $v = 5.93 \times 10^5$ m s^{-1}	
		(iii)	$\frac{hc}{\lambda} = \phi + eV_s$ $\frac{hc}{380 \times 10^{-9}} = \phi + (1.6 \times 10^{-19})(1)$ $\Phi = 2.27\ eV$	
		(iv)	Cannot be read off directly, as the stopping voltage and the wavelength do not have a linear relationship (OR they have an inverse relationship with x-intercept is not shown in the diagram)	

	(b)	**(i)**	 Fig. 5.3	
		(ii)	 • Lower intensity (than before, at all parts of new graph) • Half of original maximum photon energy • Same characteristic lines.	
		(iii)	Energy = $(1.4 - 0.27) \times 10^{-15}$ $= 1.13 \times 10^{-15}$ J = 7062 eV	
		(iv)	<u>Accelerated (or high energy) incident electron 'collides' with an electron from the innermost shell kicking it out of the *K*-shell</u>. The atom is excited due to the vacancy in the *K*-shell. An electron from the <u>M-shell transits to the vacancy</u> in the *K*-shell, <u>emitting a photon</u>.	

6	(a)	(i)	$C_? = 65\ s^{-1}$	